Geometric Context from a Single Image

Paul-Emile BARRANGER Guillaume BERNARD Côme HOSXE

GRMI - Project Proposal

1. Project Topic

Our project aims to estimate the 3D geometric structure of outdoor scenes from a single image. We will classify image regions into three main geometric classes: the ground plane, vertical surfaces, and the sky. Our work will be based on the paper "Geometric Context from a Single Image" [1]. To connect this project to the course, we will reformulate the problem using graphical models. Specifically, we plan to implement a Belief Propagation algorithm to infer the geometric labels. In addition, we will explore Graph Neural Networks (GNNs) as a modern approach for node classification. Finally, we will apply our geometric context to an object detection task: we will use it to remove false positives in face detection algorithms, such as faces incorrectly detected in the sky or on walls.

2. Algorithms

First, we will use the Felzenszwalb-Huttenlocher graph-based segmentation algorithm to group pixels into "superpixels". Then, we will extract various features for these regions, such as color, texture, and image location. For the classification step, we will implement and compare three approaches:

- An Adaboost classifier using decision trees, as described in the original paper.
- A Belief Propagation algorithm to perform a benchmark comparison.
- A Graph Neural Network (GNN) treating the superpixels as a graph. This will allow us to frame the problem as node classification, where the network learns from the context of neighboring nodes instead of relying only on hand-calculated features.

We intend to implement the core algorithms ourselves. Specifically, we will develop the Belief Propagation logic and the GNN message-passing layers from scratch. We will develop the project in Python, using OpenCV for image processing and NumPy or base PyTorch for matrix operations and auto-differentiation. But we will avoid using high-level pre-built libraries like PyTorch Geometric for the graph logic.

3. Dataset

We will use the dataset provided by the authors of the original paper, available on their website (`https://dhoiem.cs.illinois.edu/projects/data.html`). It contains 300 outdoor images that are often highly cluttered and represent a wide variety of scenes. Every image has ground truth labels for the geometric classes. We will use 50 images to train the segmentation algorithm and the remaining 250 images to train and evaluate the overall system. We may also use a standard face detection dataset to test our practical application.

4. Validation and Metrics

We will evaluate our system using 5-fold cross-validation on the 250 test images. Our primary success metric will be the pixel accuracy , which is the percentage of image pixels that have the correct geometric label for the three main classes. We will compare the pixel accuracy of our Belief Propagation and GNN implementations with the results of the original paper's algorithm. Furthermore, for our face detection application, our secondary success metric will be the reduction of false positives per image while maintaining a good detection rate.

5. References

[1] Hoiem, D. and Efros, A.A. and Hebert, M. *Geometric Context from a Single Image*. Carnegie Mellon University, 2005.